SMARTBRIDGE
Let's Bridge the Gap

Problem-Solution Fit

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

1. Customer Segment(s) [CS]	**6. Customer Limitations [CL]**	**5. Available Solutions [AS]**
Disaster Management Authorities Weather Monitoring Departments Municipal Corporations Emergency Response Teams Communities in flood-prone areas	Limited access to intelligent flood prediction tools. Dependence on manual weather analysis. Delayed interpretation of weather data.	Government weather forecasts. Rainfall monitoring reports. Manual analysis. No AI-based flood prediction.

2. Problems / Pains + Frequency [PR]	**9. Problem Root / Cause [RC]**	**7. Behavior + Intensity [BE]**
Flood warnings may be delayed during heavy rainfall. Occurs frequently during monsoon seasons.	Extreme rainfall. Changing weather patterns. Delayed weather analysis. Lack of intelligent prediction systems.	Authorities monitor rainfall and weather reports. Monitoring intensity increases during heavy rainfall.

3. Triggers to Act [TR]	**10. Your Solution [SL]**	**8. Channels of Behavior [CH]**
Continuous heavy rainfall. High humidity.	Flask web application using a trained Random Forest model to predict Flood Risk or No Flood	Online: • Web browser (Desktop/Mobile) Offline:

Increasing cloud cover. Official weather alerts.	Risk from weather and rainfall parameters.	• Disaster management offices • Emergency control rooms • Field response teams

4. Emotions Before / After [EM]
Before: Concerned, uncertain, and dependent on manual analysis during heavy rainfall. After: Better informed, confident, and prepared to make timely disaster response decisions.